

L-4

03-10-2023

~~L-4~~
~~L-4~~

From part lecture....

Show that $v = \omega \sqrt{A^2 - x^2}$

$$E_T = E_p + E_k$$

$$\Rightarrow \frac{1}{2} k A^2 = \frac{1}{2} k x^2 + \frac{1}{2} m v^2$$

$$\Rightarrow \frac{1}{2} m v^2 = \frac{1}{2} k A^2 - \frac{1}{2} k x^2$$

$$\Rightarrow m v^2 = k A^2 - k x^2$$

$$\Rightarrow m v^2 = k (A^2 - x^2)$$

$$\Rightarrow v^2 = \frac{k}{m} (A^2 - x^2)$$

$$\Rightarrow v = \sqrt{\frac{k}{m} (A^2 - x^2)}$$

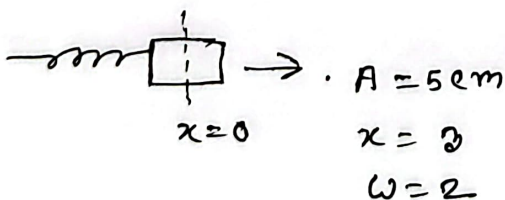
$$\Rightarrow v = \sqrt{\frac{k}{m}} \cdot \sqrt{A^2 - x^2}$$

$$\Rightarrow v = \omega \sqrt{A^2 - x^2}$$

$$\therefore \boxed{v = \omega \sqrt{A^2 - x^2}} \rightarrow \text{max time at max}$$

$$\text{or, } v = -A\omega \sin \omega t \rightarrow \text{max time at max}$$

Example:


 $x = 3$ Position $v = ?$

$$v = \omega \sqrt{A^2 - x^2}$$

$$= 2 \sqrt{5^2 - 3^2}$$

$$= 8 \text{ cm s}^{-1}$$

if given,

$$k = 19000$$

$$m = 50 \text{ g}$$

$$v = \sqrt{\frac{k}{m}} \cdot \sqrt{A^2 - x^2}$$

$$= \sqrt{\frac{19000}{50}} \cdot \sqrt{5^2 - 3^2}$$

$$= 8\sqrt{5}$$

$$= 77.97$$

if $x = 3 \cos 5t$ & $t = 2s$

$$v = \frac{dx}{dt}$$

$$= 3 \sin 5t \cdot 5$$

$$= 15 \sin 5t$$

$$= 15$$

Phase / फेज

Phase difference

$$\left(\frac{\pi}{p} + \omega\right) \text{ or } A = 5$$

अन्य संकेत
अपेक्षा

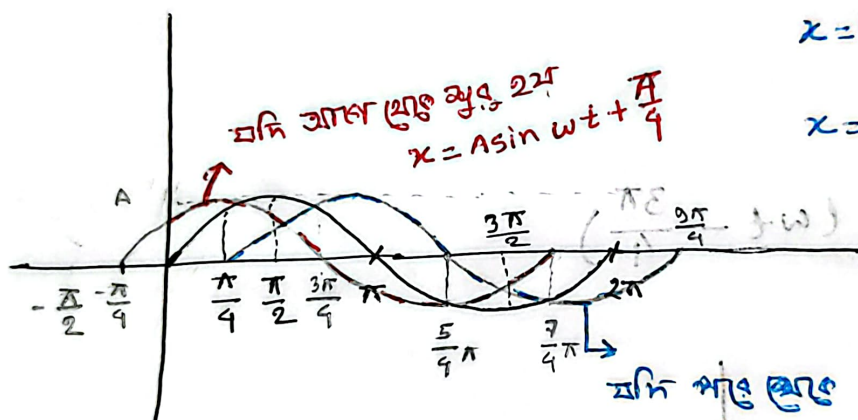
① is used to describe the phase position of one wave relative to another

$$x = A \sin \omega t$$

shifting the original graph w.r.t. to ωt using phase $[\theta]$

ज-व

displacement vs Angle graph.



$$x = A \sin \omega t + \theta \rightarrow$$

$$x = A \sin \omega t - \theta$$

middle point जहाँ
way $\frac{1s + last}{2}$
**

अन्य संकेत

अपेक्षा

$$x = A \sin \omega t - \frac{\pi}{4}$$

way 2

① value तर्क 0 जहाँ जहाँ जहाँ

② $\omega t + x$ or $\omega t - x$ जहाँ angle जहाँ जहाँ

③ '+' जहाँ left

④ '-' जहाँ Right

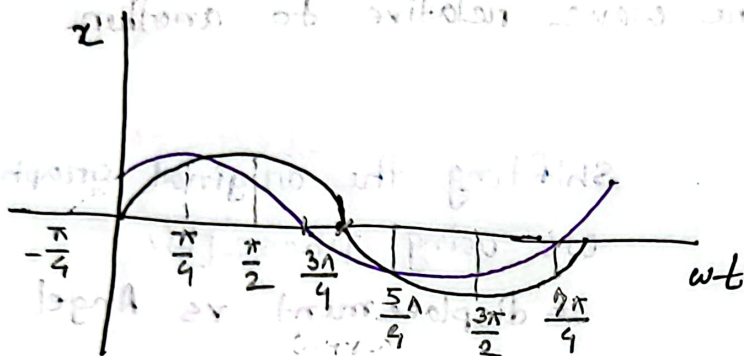
Question

① $x = A \sin(\omega t + \frac{\pi}{4})$

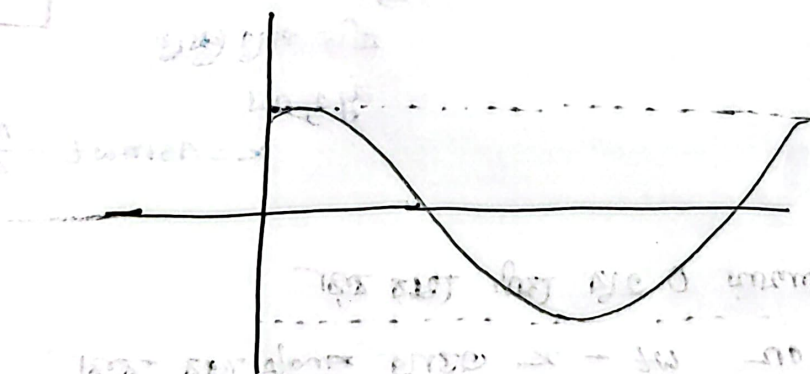
$\omega t + \frac{\pi}{4} = 0$

$\omega t = -\frac{\pi}{4}$

Construct new graph at



② $x = A \cos(\omega t - \frac{3\pi}{4})$

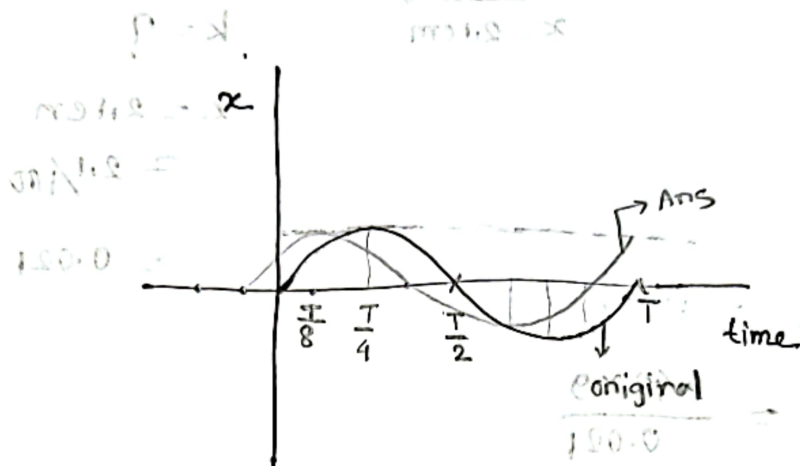


Let $x = A \sin(\omega t + \phi)$ (iii)

Let $x = A \cos(\omega t + \phi)$ (iv)

Time

$$x = A \sin(\omega t + \frac{\pi}{4})$$



$$\omega t = -\frac{\pi}{4}$$

$$\frac{2\pi}{T} t = -\frac{\pi}{4}$$

$$t = -\frac{\pi}{4} \times \frac{T}{2\pi}$$

$$= -\frac{T\pi}{8\pi}$$

$$= -\frac{T}{8}$$

Start here

$$x = A \sin(\omega t - \frac{\pi}{2})$$

$$k = 10.6 \text{ N/m}$$

$$A = 0.100 \text{ m}$$

$$x = 0.100 \sin(8.08t)$$

$$v = 0.808 \sin(8.08t)$$

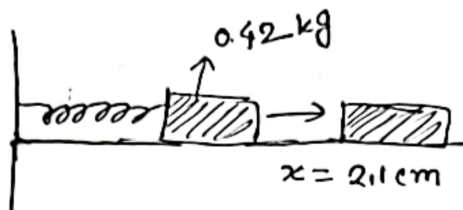
$$(x, v, a)$$

$$v = 0.808 \sin(8.08t)$$

$$v = 0.808 \sin(8.08t)$$

Meth.

①



$$m = 0.42 \text{ kg}$$

$$a = 9.0 \text{ ms}^{-2}$$

$$k = ?$$

$$x = 2.1 \text{ cm}$$

$$= 2.1/100$$

$$= 0.021$$

$$F = kx$$

$$k = \frac{F}{x}$$

$$= \frac{ma}{x} = \frac{0.42 \times 9}{0.021}$$

$$= 180 \text{ N/m}$$

②

$$k = 19.6 \text{ N/m}$$

$$A = 0.100 \text{ m}$$

$$x = -0.1 \cos 8.08t$$

$$v = 0.808 \sin 8.08t$$

a) total energy?

b) E_k and E_p as a function of time

c) the velocity when the mass is 0.050 m from eqb

d) E_k and E_p at half Amplitude ($x = \pm A/2$)Soln

$$a) E_T = \frac{1}{2} k A^2$$

$$= \frac{1}{2} \times 19.6 \times (0.100)^2$$



$$\textcircled{b} E_k = \frac{1}{2} m v^2$$

$$= \frac{1}{2} \times m (0.008 \sin 8.08t)^2$$

$$= \frac{1}{2} m (0.008)^2 \sin^2 8.08t$$

$$E_p = \frac{1}{2} k x^2$$

$$= \frac{1}{2} \times 19.6 \times (-0.1 \cos 8.08t)^2$$

$$= \frac{1}{2} \times 19.6 \times (-0.1)^2 \cos^2 8.08t$$

$$\textcircled{c} A = 0.1 \text{ m}$$

$$x = 0.05 \text{ m}$$

$$x = -0.1 \cos 8.08t$$

$$\therefore \omega = 8.08$$

$$\therefore v = \omega \sqrt{A^2 - x^2}$$

$$= 8.08 \sqrt{(0.1)^2 - (0.05)^2}$$

$$= 0.60$$

$$\textcircled{d} E_T = \frac{1}{2} k A^2$$

$$E_p = \frac{1}{2} k x^2$$

$$= \frac{1}{2} k \left(\frac{A}{2}\right)^2$$

$$= \frac{1}{2} k \cdot \frac{A^2}{4}$$

$$= \frac{1}{4} \cdot \frac{1}{2} k A^2$$

$$= \frac{1}{4} \cdot E_T$$

$$E_k = E_T - E_p$$

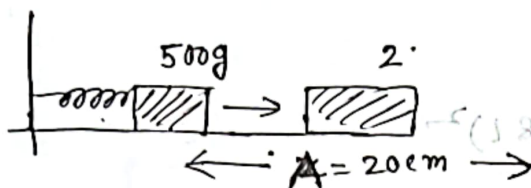
$$\therefore E_k = E_T - E_p$$

$$= E_T - \frac{1}{4} E_T = \frac{1}{2} k A^2 - \frac{1}{4} k A^2$$

$$= \frac{4E_T - E_T}{4}$$

$$= \frac{3}{4} E_T$$

③



$$A = 20 \text{ cm} = 0.2 \text{ m}$$

$$m = 500 \text{ g} = 0.5 \text{ kg}$$

$$v = \omega \sqrt{A^2 - x^2}$$

$$T = 0.8 \text{ s}$$

$$\Rightarrow v = \frac{2\pi}{T} \sqrt{A^2 - x^2}$$

$$v = 1.0 \text{ m s}^{-1}$$

$$\Rightarrow \frac{v^2}{\left(\frac{2\pi}{T}\right)^2} = A^2 - x^2$$

$$\Rightarrow x^2 = A^2 - \frac{v^2}{\left(\frac{2\pi}{T}\right)^2}$$

$$\Rightarrow x = \pm \sqrt{A^2 - \frac{v^2}{\left(\frac{2\pi}{T}\right)^2}}$$

$$= \pm \sqrt{(0.2)^2 - \frac{(1)^2}{\left(\frac{2\pi}{0.8}\right)^2}}$$

$$x = \pm 0.154 \cdot A \text{ m}$$

④

$$x = A \sin \omega t$$

$$v = \frac{dx}{dt}$$

$$= A \cos \omega t \cdot \omega$$

$$= A \omega \cos \omega t$$

$$\therefore v_{\text{max}} = A \omega$$

$$= (0.2 \times 10^{-3}) \times 2\pi \times 1 \times 10^3$$

$$= 1.256 \text{ m s}^{-1}$$

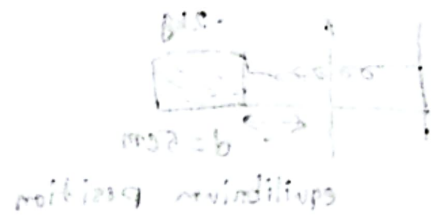
$$A = 0.2 \text{ mm}$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$f = 1 \text{ kHz} = 1 \times 10^3 \text{ Hz}$$

⑥

100 MHz
 $R = 2 \text{ k}\Omega$



⑦

$$\omega = \sqrt{\frac{1}{LC}} = \frac{1}{\sqrt{LC}} = \omega$$

$$\frac{\pi R}{T} = \omega = \omega$$

$$\omega = \omega$$

$$2\pi f \cdot 1 = \frac{\omega R}{\pi R} \rightarrow \frac{\omega}{\pi R} = \frac{1}{2\pi}$$

⑤ $f = 100 \text{ MHz}$
 $= 1 \times 10^8 \text{ Hz}$

$$\frac{\pi R}{\omega} = T$$

$$T = ?$$

$$\frac{\pi R}{\omega} = T$$

$$T = \frac{1}{f} = \frac{1}{1 \times 10^8}$$

$$= 1 \times 10^{-8} \text{ s}$$

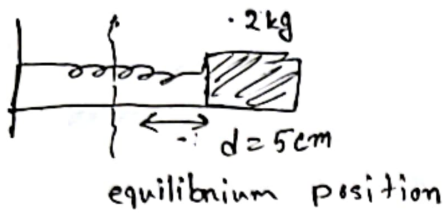
$$T = 1 \times 10^{-8} \text{ s}$$

Amplitude and angular velocity of the signal

⑥ Amplitude and angular velocity of the signal

Amplitude
 II

7



$$k = 196 \text{ N/m}$$

$$m = 2 \text{ kg}$$

d

$$\omega = \sqrt{\frac{k}{m}} = \sqrt{\frac{196}{2}} = 7\sqrt{2}$$

$$\omega = 2\pi f$$

$$\omega = 2\pi f = \frac{2\pi}{T}$$

$$f = \frac{\omega}{2\pi} = \frac{7\sqrt{2}}{2\pi} = 1.575$$

$$T = \frac{2\pi}{\omega}$$

$$= \frac{2\pi}{7\sqrt{2}}$$

$$= 0.639$$

SHM 001

SH 001

C = T

$$\frac{1}{2\pi \times 1} - \frac{1}{\gamma} = T$$

$$2 \times 10^{-1} \times 1 =$$

b

$$x = 5 \cos 7\sqrt{2}t$$

released at $t=0$ s. so, initial Phase/0 means in

position when released is like simple harmonic motion

SHM

8

a) 15 completed at $\cdot 10 \text{ s}$ / or $T = \frac{1}{f}$
 $1 \quad \quad \quad \frac{10}{15} \text{ s}$
 $= 0.667$

b)

$$V_{\max} = \omega A$$

$$A = 2 \text{ cm.}$$

$$= 0.2 \text{ m}$$

$$= \frac{2\pi}{T} \times 0.2$$

$$= 1.88 \text{ ms}^{-1}$$

c)

$$\omega = \frac{2\pi}{T} = 9.38$$

$$x = A \cos \omega t$$

$$= 0.2 \cos 9.38 t$$

$$v = -A\omega \sin \omega t$$

$$= -(0.2 \times 9.38) \sin 9.38 t$$

$$= -1.88 \sin 9.38 t$$

$$= -1.79 \text{ ms}^{-1}$$

$$t = 0.8 \text{ s}$$

calculator $\rightarrow$ Radian

mode $\rightarrow$ calculate

answer